BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Practical No: 4

Division: C Batch: 1

Roll No: 16 PRN No: 12412535

Name of Student: Abhishek Dasharath Kadam

Subject Name: Deep Learning

Problem Statement:

Develop an LSTM-based model for time-series forecasting using stock price, weather, or sales datasets.

Algorithm :

Theory

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network designed for sequential and time-series data. LSTM uses memory cells and gates to retain important information from previous time steps. This makes it suitable for forecasting problems where the current value depends on previous observations.

In this experiment, historical weather observations are provided to an LSTM model. The model uses the previous 24 hourly observations to predict the temperature for the next hour.

Dataset Description

The Jena Climate dataset contains weather observations collected at the Max Planck Institute for Biogeochemistry weather station in Jena, Germany. The original data contains 14 weather variables recorded every 10 minutes over several years.

For this experiment, the data is resampled from 10-minute observations to hourly observations and the following variables are used:

Feature	Description
p (mbar)	Atmospheric pressure
T (degC)	Air temperature
Tpot (K)	Potential temperature
Tdew (degC)	Dew-point temperature
rh (%)	Relative humidity
wv (m/s)	Wind velocity
max. wv (m/s)	Maximum wind velocity

The target variable is T (degC).

Methodology

Jena Climate Dataset
↓
Data Cleaning
↓
Convert Date-Time
↓
Hourly Resampling
↓
Feature Selection
↓
80% Training / 20% Testing
↓
Min-Max Normalization
↓
24-Hour Sequence Creation
↓
LSTM Model
↓
Model Training
↓
Temperature Prediction
↓
MSE / RMSE / MAE / R^2
↓
Actual vs Predicted Graph

LSTM

LSTM stands for Long Short-Term Memory.
It is a type of Recurrent Neural Network designed to learn long-term dependencies in sequential data.
An LSTM cell consists mainly of:

1. Forget gate
2. Input gate
3. Cell state
4. Output gate

1. Forget Gate

The forget gate determines which information from the previous cell state should be discarded.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

where:

- f_t= forget gate
- x_t= current input
- h_{t-1}= previous hidden state
- σ= sigmoid activation

2. Input Gate

The input gate determines which new information should be stored.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

A candidate cell state is calculated as:

$$C_t = \tanh(W_C[h_{t-1}, x_t] + b_C)$$

3. Cell State

The cell state is updated using the forget and input gates:

$$C_t = f_t * C_{t-1} + i_t * C_t$$

It acts as the memory of the LSTM.

4. Output Gate

The output gate determines what information is passed to the next layer.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

The hidden state is:

$$h_t = o_t * \tanh(C_t)$$

Thus, LSTM can selectively remember and forget information while processing sequential data.

Code :

```
import os
import zipfile
import urllib.request

import numpy as np
import pandas as pd
```

```
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

# 1. DOWNLOAD DATASET
url  =  "https://storage.googleapis.com/tensorflow/tf-keras-datasets/jena_climate_2009_2016.csv.zip"

zip_path = "/content/jena_climate.zip"
extract_path = "/content/jena_climate"

if not os.path.exists(zip_path):
    urllib.request.urlretrieve(url, zip_path)

if not os.path.exists(extract_path):
    os.makedirs(extract_path)

with zipfile.ZipFile(zip_path, "r") as zip_ref:
    zip_ref.extractall(extract_path)

# 2. FIND CSV FILE
csv_file = None

for root, dirs, files in os.walk(extract_path):
    for file in files:
        if file.endswith(".csv"):
            csv_file = os.path.join(root, file)
            break

if csv_file is None:
    raise FileNotFoundError("CSV file not found.")

print("Dataset:", csv_file)

# 3. LOAD DATASET
df = pd.read_csv(csv_file)

print("\nDataset Shape:")
print(df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nFirst 5 Rows:")
print(df.head())
```

```
# 4. DATE-TIME PROCESSING
df["Date Time"] = pd.to_datetime(df["Date Time"])

df = df.set_index("Date Time")

df = df.sort_index()

# 5. SELECT FEATURES
features = [
    "p (mbar)",
    "T (degC)",
    "Tpot (K)",
    "Tdew (degC)",
    "rh (%)",
    "wv (m/s)",
    "max. wv (m/s)"
]

data = df[features].copy()

# 6. HANDLE MISSING VALUES
data = data.interpolate()

data = data.dropna()

# 7. RESAMPLE INTO HOURLY DATA
data = data.resample("1h").mean()

data = data.interpolate()

data = data.dropna()


print("\nHourly Dataset Shape:")
print(data.shape)

print("\nHourly Data:")
print(data.head())

# 8. VISUALIZE TEMPERATURE
plt.figure(figsize=(14,  5))

plt.plot(data.index, data["T (degC)"])

plt.title("Hourly Temperature")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
```

```
plt.grid(True)
plt.show()

# 9. TRAIN-TEST SPLIT
split_index = int(len(data) * 0.80)

train_data = data.iloc[:split_index]
test_data = data.iloc[split_index:]


print("\nTraining Samples:", len(train_data))
print("Testing Samples:", len(test_data))

# 10. SCALE DATA
scaler = MinMaxScaler()

train_scaled = scaler.fit_transform(train_data)

test_scaled = scaler.transform(test_data)

# 11. CREATE SEQUENCES
lookback = 24

target_index = features.index("T (degC)")


def create_sequences(dataset, lookback):
    X = []
    y = []

    for i in range(lookback, len(dataset)):
        X.append(dataset[i-lookback:i])
        y.append(dataset[i, target_index])

    return np.array(X), np.array(y)


X_train, y_train = create_sequences(train_scaled, lookback)

X_test, y_test = create_sequences(test_scaled, lookback)


print("\nX_train Shape:", X_train.shape)
print("y_train Shape:", y_train.shape)

print("X_test Shape:", X_test.shape)
print("y_test Shape:", y_test.shape)
# 12. BUILD LSTM MODEL
model = Sequential()
```

```
model.add(
   LSTM(
      64,
      return_sequences=True,
      input_shape=(X_train.shape[1], X_train.shape[2])
   )
)

model.add(Dropout(0.2))

model.add(LSTM(32))

model.add(Dropout(0.2))

model.add(Dense(16, activation="relu"))

model.add(Dense(1))

# 13. COMPILE MODEL
model.compile(
   optimizer="adam",
   loss="mse",
   metrics=["mae"]
)

print("\nModel Summary:")
model.summary()

# 14. EARLY STOPPING
early_stop = EarlyStopping(
   monitor="val_loss",
   patience=5,
   restore_best_weights=True
)

# 15. TRAIN MODEL
history = model.fit(
   X_train,
   y_train,
   epochs=30,
   batch_size=64,
   validation_split=0.1,
   callbacks=[early_stop],
   shuffle=False,
   verbose=1
)

# 16. PREDICTION
```

```
y_pred_scaled = model.predict(X_test)

# 17. CONVERT BACK TO ORIGINAL TEMPERATURE
dummy_actual = np.zeros(
    (len(y_test), len(features))
)

dummy_pred = np.zeros(
    (len(y_pred_scaled), len(features))
)

dummy_actual[:, target_index] = y_test

dummy_pred[:, target_index] = y_pred_scaled[:, 0]


y_test_actual = scaler.inverse_transform(dummy_actual)[:, target_index]

y_pred_actual = scaler.inverse_transform(dummy_pred)[:, target_index]

# 18. CALCULATE METRICS
mse = mean_squared_error(
    y_test_actual,
    y_pred_actual
)

rmse = np.sqrt(mse)

mae = mean_absolute_error(
    y_test_actual,
    y_pred_actual
)

r2 = r2_score(
    y_test_actual,
    y_pred_actual
)

print("MODEL EVALUATION")

print("MSE  :", mse)
print("RMSE :", rmse)
print("MAE  :", mae)
print("R²   :", r2)


# 19. PLOT ACTUAL VS PREDICTED

plt.figure(figsize=(15, 6))
```

```
plt.plot(
    y_test_actual[:500],
    label="Actual Temperature"
)

plt.plot(
    y_pred_actual[:500],
    label="Predicted Temperature"
)

plt.title("Actual vs Predicted Temperature")

plt.xlabel("Test Time Steps")

plt.ylabel("Temperature (°C)")

plt.legend()

plt.grid(True)

plt.show()

# 20. TRAINING AND VALIDATION LOSS
plt.figure(figsize=(10,  5))

plt.plot(
    history.history["loss"],
    label="Training Loss"
)

plt.plot(
    history.history["val_loss"],
    label="Validation Loss"
)

plt.title("Training and Validation Loss")

plt.xlabel("Epoch")

plt.ylabel("MSE Loss")

plt.legend()

plt.grid(True)

plt.show()

# 21. PREDICT NEXT HOUR
```

```
last_sequence = test_scaled[-lookback:]

last_sequence = last_sequence.reshape(
    1,
    lookback,
    len(features)
)

next_temp_scaled = model.predict(last_sequence)

dummy_next = np.zeros(
    (1, len(features))
)

dummy_next[0, target_index] = next_temp_scaled[0, 0]

next_temperature = scaler.inverse_transform(
    dummy_next
)[0, target_index]

print("NEXT HOUR TEMPERATURE")

print(
    "Predicted Temperature:",
    round(float(next_temperature), 2),
    "°C"
)
```

Output :

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/jena_climate_2009_2016.csv.zip
13568290/13568290 ━━━━━━━━━━━━━━━━━━━━ **0s** 0us/step
Dataset Shape: (420551, 15)

First 5 Rows:

	Date Time	p (mbar)	T (degC)	Tpot (K)	Tdew (degC)	rh (%)	\
0	01.01.2009 00:10:00	996.52	-8.02	265.40	-8.90	93.3	
1	01.01.2009 00:20:00	996.57	-8.41	265.01	-9.28	93.4	
2	01.01.2009 00:30:00	996.53	-8.51	264.91	-9.31	93.9	
3	01.01.2009 00:40:00	996.51	-8.31	265.12	-9.07	94.2	

```
4  01.01.2009 00:50:00    996.51    -8.27    265.15       -9.04    94.1

   VPmax (mbar)  VPact (mbar)  VPdef (mbar)  sh (g/kg)  H2OC (mmol/mol)  \
0          3.33          3.11          0.22       1.94             3.12
1          3.23          3.02          0.21       1.89             3.03
2          3.21          3.01          0.20       1.88             3.02
3          3.26          3.07          0.19       1.92             3.08
4          3.27          3.08          0.19       1.92             3.09

   rho (g/m**3)  wv (m/s)  max. wv (m/s)  wd (deg)
0       1307.75      1.03           1.75     152.3
1       1309.80      0.72           1.50     136.1
2       1310.24      0.19           0.63     171.6
3       1309.19      0.34           0.50     198.0
4       1309.00      0.32           0.63     214.3

Columns:
['Date Time', 'p (mbar)', 'T (degC)', 'Tpot (K)', 'Tdew (degC)', 'rh (%)', 'VPmax (mbar)', 'VPact
(mbar)', 'VPdef (mbar)', 'sh (g/kg)', 'H2OC (mmol/mol)', 'rho (g/m**3)', 'wv (m/s)', 'max. wv (m/s)',
'wd (deg)']



          Date Time  p (mbar)  T (degC)  Tpot (K)  Tdew (degC)  rh (%)  \
0 2009-01-01 00:10:00    996.52     -8.02    265.40        -8.90    93.3
1 2009-01-01 00:20:00    996.57     -8.41    265.01        -9.28    93.4
2 2009-01-01 00:30:00    996.53     -8.51    264.91        -9.31    93.9
3 2009-01-01 00:40:00    996.51     -8.31    265.12        -9.07    94.2
4 2009-01-01 00:50:00    996.51     -8.27    265.15        -9.04    94.1

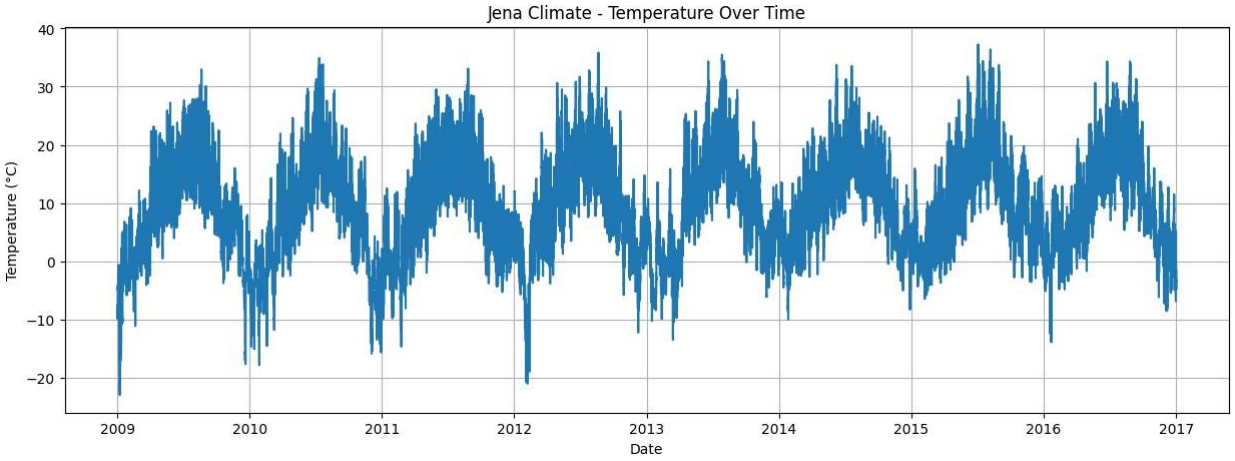
   VPmax (mbar)  VPact (mbar)  VPdef (mbar)  sh (g/kg)  H2OC (mmol/mol)  \
0          3.33          3.11          0.22       1.94             3.12
1          3.23          3.02          0.21       1.89             3.03
2          3.21          3.01          0.20       1.88             3.02
3          3.26          3.07          0.19       1.92             3.08
4          3.27          3.08          0.19       1.92             3.09

   rho (g/m**3)  wv (m/s)  max. wv (m/s)  wd (deg)
0       1307.75      1.03           1.75     152.3
1       1309.80      0.72           1.50     136.1
2       1310.24      0.19           0.63     171.6
3       1309.19      0.34           0.50     198.0
4       1309.00      0.32           0.63     214.3
          Date Time  p (mbar)  T (degC)  Tpot (K)  Tdew (degC)  rh (%)  \
```

420546	2016-12-31 23:20:00	1000.07	-4.05	269.10	-8.13	73.10
420547	2016-12-31 23:30:00	999.93	-3.35	269.81	-8.06	69.71
420548	2016-12-31 23:40:00	999.82	-3.16	270.01	-8.21	67.91
420549	2016-12-31 23:50:00	999.81	-4.23	268.94	-8.53	71.80
420550	2017-01-01 00:00:00	999.82	-4.82	268.36	-8.42	75.70

	VPmax (mbar)	VPact (mbar)	VPdef (mbar)	sh (g/kg)	H2OC (mmol/mol)	\
420546	4.52	3.30	1.22	2.06	3.30	
420547	4.77	3.32	1.44	2.07	3.32	
420548	4.84	3.28	1.55	2.05	3.28	
420549	4.46	3.20	1.26	1.99	3.20	
420550	4.27	3.23	1.04	2.01	3.23	

	rho (g/m**3)	wv (m/s)	max. wv (m/s)	wd (deg)
420546	1292.98	0.67	1.52	240.0
420547	1289.44	1.14	1.92	234.3
420548	1288.39	1.08	2.00	215.2
420549	1293.56	1.49	2.16	225.8
420550	1296.38	1.23	1.96	184.9



Hourly Dataset Shape: (70129, 14)

	p (mbar)	T (degC)	Tpot (K)	Tdew (degC)	rh (%)	VPmax (mbar)	VPact (mbar)	VPdef (mbar)	sh (g/kg)	H2OC (mmol/mol)	rho (g/m**3)	wv (m/s)	max. wv (m/s)	wd (deg)
Date Time														
2009-01-01 00:00:00	996.528000	-8.304000	265.118000	-9.120000	93.780000	3.260000	3.058000	0.202000	1.910000	3.068000	1309.196000	0.520000	1.020000	174.460000
2009-01-01 01:00:00	996.525000	-8.065000	265.361667	-8.861667	93.933333	3.323333	3.121667	0.201667	1.951667	3.133333	1307.981667	0.316667	0.711667	172.416667
2009-01-01 02:00:00	996.745000	-8.763333	264.645000	-9.610000	93.533333	3.145000	2.940000	0.201667	1.836667	2.950000	1311.816667	0.248333	0.606667	196.816667
2009-01-01 03:00:00	996.986667	-8.896667	264.491667	-9.786667	93.200000	3.111667	2.898333	0.210000	1.811667	2.906667	1312.813333	0.176667	0.606667	157.083333
2009-01-01 04:00:00	997.158333	-9.348333	264.026667	-10.345000	92.383333	3.001667	2.775000	0.231667	1.733333	2.780000	1315.355000	0.290000	0.670000	150.093333

Final Dataset Shape: (70041, 7)

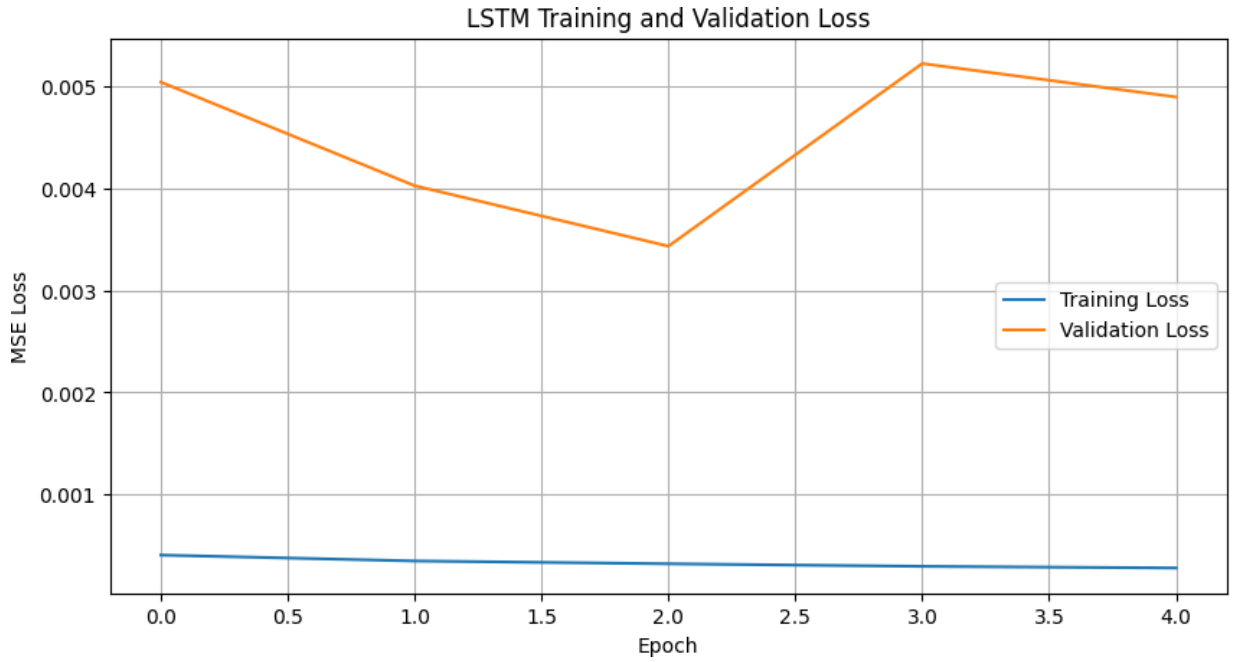
Training Samples: 56032
Testing Samples: 14009

X_train: (56008, 24, 7)
y_train: (56008,)
X_test: (13985, 24, 7)
y_test: (13985,)

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 24, 64)	18,432
dropout (Dropout)	(None, 24, 64)	0
lstm_1 (LSTM)	(None, 32)	12,416
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 16)	528
dense_1 (Dense)	(None, 1)	17

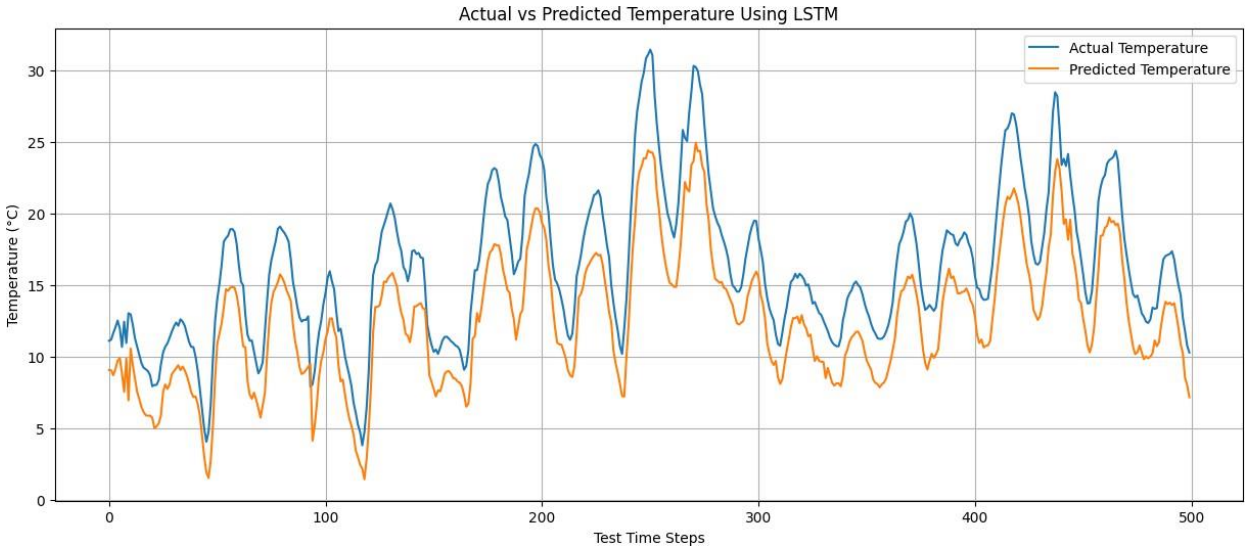
Total params: 31,393 (122.63 KB)
Trainable params: 31,393 (122.63 KB)
Non-trainable params: 0 (0.00 B)

Epoch 1/5
876/876 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **30s** 35ms/step - loss: 4.0734e-04 - mae: 0.0152 - val_loss: 0.0050 - val_mae: 0.0670
Epoch 2/5
876/876 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **29s** 33ms/step - loss: 3.4958e-04 - mae: 0.0140 - val_loss: 0.0040 - val_mae: 0.0594
Epoch 3/5
876/876 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **29s** 33ms/step - loss: 3.2214e-04 - mae: 0.0134 - val_loss: 0.0034 - val_mae: 0.0539
Epoch 4/5
876/876 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **40s** 32ms/step - loss: 2.9700e-04 - mae: 0.0128 - val_loss: 0.0052 - val_mae: 0.0654
Epoch 5/5
876/876 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **27s** 31ms/step - loss: 2.8117e-04 - mae: 0.0125 - val_loss: 0.0049 - val_mae: 0.0624



438/438 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ **3s** 7ms/step

==========
LSTM MODEL PERFORMANCE
==========
MSE : 11.6135
RMSE : 3.4079
MAE : 3.1376
R^2 : 0.8228



1/1 ——————————————————————— **0s** 41ms/step

Predicted Temperature for Next Hour: -6.71 °C